

The Dobbertin Story, in Pictures

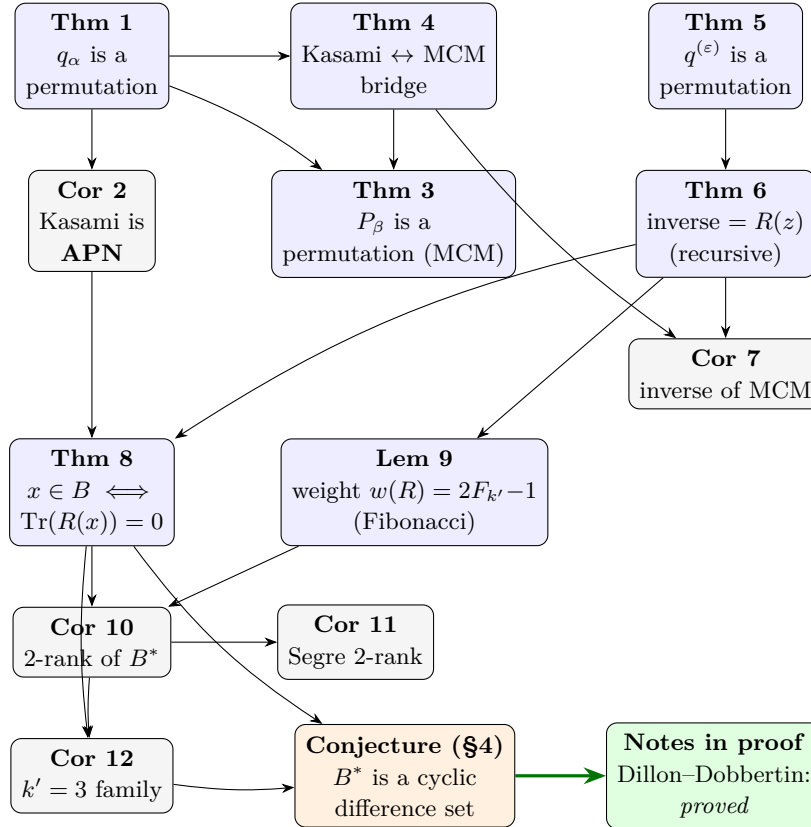
Kasami power functions, permutation polynomials, and cyclic difference sets
which structural patterns appear, and how they travel through the theorems

Abstract

One paper, six recurring *shapes*. Each theorem, corollary, and the final conjecture is one of these shapes wearing a new costume. This is a picture book: little prose, many arrows. We name each pattern, draw it, point to where it lives in Dobbertin (1999), and note how it is mirrored in the Lean project.

$$d = 2^{2k} - 2^k + 1 \text{ (Kasami exponent), } \gcd(k, n) = 1, \quad k' \equiv 1/k \pmod{n}, \quad \text{field } L = \text{GF}(2^n) = \mathbb{F}_{2^n}.$$

The whole paper on one page



■ theorem ■ corollary ■ conjecture ■ later proved

★ BIG IDEA

Read left to right. **Permutations** give **APN**; the **bridge** copies everything between two worlds; **inverting** a permutation gives an explicit polynomial R ; R gives a **trace test** for a set B ; the **recursion** behind R is **Fibonacci**, which fixes the **2-rank**; the trace test finally powers a **difference-set** conjecture.

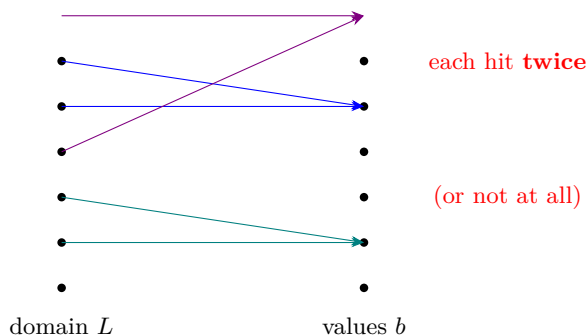
The six patterns at a glance

#	pattern (simple English)	En- where in Dobbertin	Lean mirror
P1	Two-to-one / width 2: a map hits each value exactly 0 or 2 times	Cor 2 (APN); two-to-one map $t^{2^k} + t$	<code>wid</code> , <code>WidthBdd</code>
P2	Bridge: relabel by an invertible map, properties copy across	Thm 4 (Kasami $\leftrightarrow$ MCM), Thm 3	<code>SiteEquiv</code> , <code>transport</code>
P3	Clock arithmetic: exponents live on a circle of size $2^n - 1$	reduced normal form, cyclic cosets	<code>pow_invariant_mod</code>
P4	Fibonacci growth: a two-step recursion counts the terms	Lem 9, $w(R) = 2F_{k'} - 1$; Cor 10/12	<code>recursion</code> & <code>Nat.fib</code>
P5	Quadratic echo: whatever $2^k + 1$ does, $2^{2k} - 2^k + 1$ echoes	Introduction; Cor 11; §4	Kasami vs. Gold
P6	Flat / ideal: a count is perfectly even $\Rightarrow$ difference set	§4 conjecture (ideal autocorrelation)	<code>WidthUnif</code> , <code>flat spectrum</code>

1 P1 — Two-to-one (= width 2 = APN)

Plain words: the difference map lands on each target either 0 times or exactly 2 times.

$x \mapsto (x + a)^d + x^d = b$ has 0 or 2 solutions.



IN THE PAPER

Cor 2: “Kasami power functions are almost perfect nonlinear.” The engine is a permutation q (Thm 1) together with the two-to-one map $t \mapsto t^{2^k} + t$, so $p(t) = \frac{1}{\Delta} q(t^{2^k} + t)$ is exactly 2-to-1.

✦ THE PATTERN

“2 solutions” is the number that everything else is measured against: $width = \text{size of a fibre}$.
 $APN = width \leq 2$.

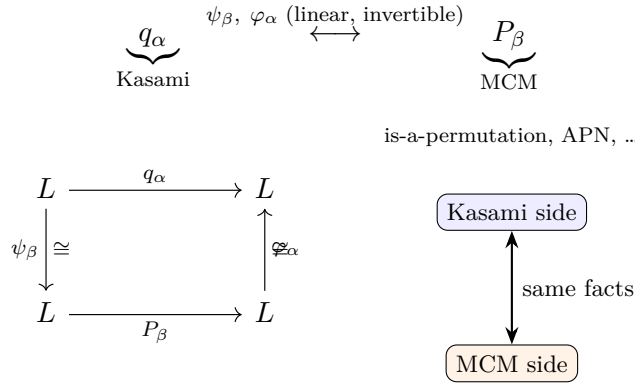
✿ IN THE LEAN PROJECT

The project’s EWIGO layer makes this the primitive object: the twisted difference `tDiff`, its fibre `fib`, and the counter `wid`. “APN” is just `WidthBdd _ 2`.

```
def tDiff (d : ℕ) (a x : S.obj) : S.obj :=
  S.prod_ (S.endo d (S.prod_ x a)) (S.inv_ (S.endo d x))
def wid (d : ℕ) (a b : S.obj) : ℕ := (fib S d a b).card
def WidthBdd ω (d : ℕ) : Prop :=
  ∀ a, a ≠ S.unit_ → ∀ b, wid S d a b ≤ ω
```

2 P2 — The Bridge (transport along an invertible relabelling)

Plain words: two families look different but a reversible dictionary turns one into the other, so a proof on one side is free on the other.



✎ IN THE PAPER

Thm 4: an explicit *linearized* substitution $\psi_\beta(x) = \sum_{i < k} x^{2^i} + \beta \text{Tr}(x)$ (and its inverse φ_α) gives a one-to-one correspondence $q_\alpha \circ \psi_\beta = P_\beta$. Thm 3 (MCM is a permutation) then falls out of Thm 1 for free. Dobbertin’s own words: the Kasami polynomials are “not new” — the bridge reveals they were MCM all along.

✦ THE PATTERN

The dictionary is an **additive bijection** (a linearized polynomial). Any property that only cares about the additive structure — being a permutation, being APN — is **transported** unchanged.

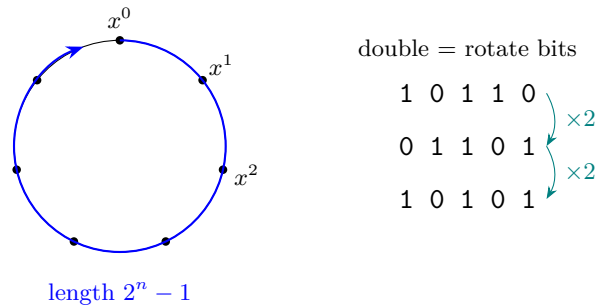
✧ IN THE LEAN PROJECT

This is exactly the “Caramello bridge” formalised in `Kasami/SiteEquivalences.lean`: `Pattern.transport` moves the collision identity along a ring iso, `SiteEquiv` makes “same picture” an equivalence relation, and `width_transport` proves the width count is literally identical on both sides.

3 P3 — Clock arithmetic on exponents (cyclotomic cosets)

Plain words: for $x \neq 0$ the exponent only matters modulo $2^n - 1$, and doubling an exponent (Frobenius $x \mapsto x^2$) just rotates its bit pattern.

$$x^e = x^{e \bmod (2^n - 1)}, \quad e \sim 2e \sim 4e \sim \dots \pmod{2^n - 1} \quad (\text{a cyclotomic coset}).$$



✎ IN THE PAPER

Used to build the *reduced normal form* of R : reduce exponents mod $2^n - 1$, then cancel cyclotomic-equivalent powers (they give the same trace). The whole rigidity argument in Lem 9 — “two exponents can’t match by a cyclic shift” — lives on this clock.

✦ THE PATTERN

Frobenius = “rotate the bits”. Trace is blind to rotation. So the true objects are **necklaces** of bits, not numbers.

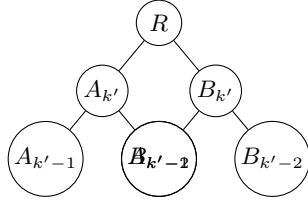
✧ IN THE LEAN PROJECT

The atom `pow_invariant_mod` ($a \equiv b \Rightarrow x^a = x^b$ for $x \neq 0$) and the Frobenius bijection `frob_bij` are the formal shadows of this clock.

4 P4 — Fibonacci growth (a two-step recursion counts terms)

Plain words: the pieces of the inverse polynomial R are built by “new = previous two”, so their term-counts are Fibonacci numbers.

$$A_{i+2} = (\cdots)A_{i+1} + (\cdots)A_i, \quad w(A_i) = F_{i-1}, \quad w(B_i) = F_{i-2} \implies \boxed{w(R) = 2F_{k'} - 1}$$



i	1	2	3	4	5	6	7
$w(A_i) = F_{i-1}$	1	1	2	3	5	8	13

✎ IN THE PAPER

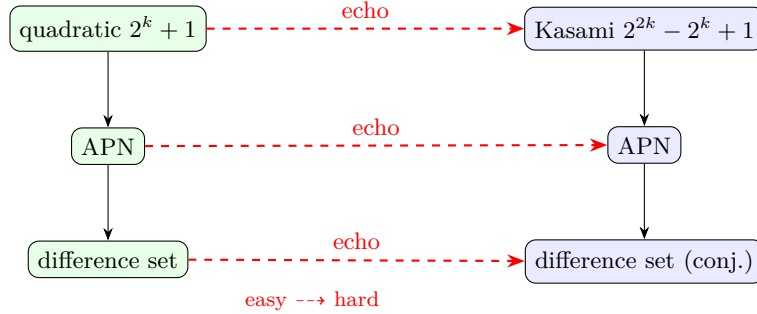
Lem 9: R is reduced iff $k' < n/2$, and $w(R) = 2F_{k'} - 1$ with $F_0 = F_1 = 1$, $F_{i+2} = F_{i+1} + F_i$. The Fibonacci count is exponential in k' yet tiny compared to a random permutation polynomial — that sparsity is the whole point.

✦ THE PATTERN

Count first, evaluate never. The 2-rank / linear span $n(2F_{k_i} - 1) + 1$ (Cor 10) is **Fibonacci wearing a linear-algebra hat**; Cor 11 reads off the Segre 2-rank as the special case $k = 2$.

5 P5 — The Quadratic Echo ($2^k + 1 \rightsquigarrow 2^{2k} - 2^k + 1$)

Plain words: the easy quadratic exponent $2^k + 1$ has nice properties; the harder Kasami exponent $2^{2k} - 2^k + 1$ has the *same* properties — just much harder to prove.



✎ IN THE PAPER

Dobbertin, Introduction: “It is an amazing observation that useful properties of the quadratic exponent $2^k + 1$ are often also shared by the Kasami exponent $2^{2k} - 2^k + 1$... but the known proofs for Kasami exponents require much more effort.” In §4 he even conjectures these are (up to equivalence) the *only* exponents giving the difference-set property.

✦ THE PATTERN

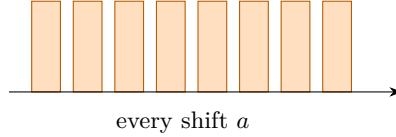
This meta-pattern is why the paper exists: transport the easy story to the hard exponent. It is P2 (the bridge) applied to a whole *theory*, not a single map.

6 P6 — Flat / Ideal (perfectly even counts $\Rightarrow$ difference set)

Plain words: if a shifted-correlation count is the *same* for every nonzero shift, the set is a (cyclic) difference set.

$$\sum_{x \in L} (-1)^{\text{Tr}(R(x) + R(ax))} = 0 \quad \text{for all } a \neq 1 \iff B^* \text{ is a difference set.}$$

ideal autocorrelation: all counts equal



IN THE PAPER

§4 Conjecture: B^* is a cyclic difference set with the *Singer parameters* $(2^n - 1, 2^{n-1} - 1, 2^{n-2} - 1)$. Special cases already known: $k = 2$ gives Segre difference sets; $k' = 2$ gives the Gong–Gaal–Golomb set; $k' = 3$ relates to the No–Chung–Yun conjecture. **Notes added in proof:** proved by Dillon & Dobbertin — all listed conjectures confirmed.

♦ THE PATTERN

“Flat” is the second-moment twin of P1’s “width 2”. A uniform width and a flat spectrum are the same fact seen from two sides.

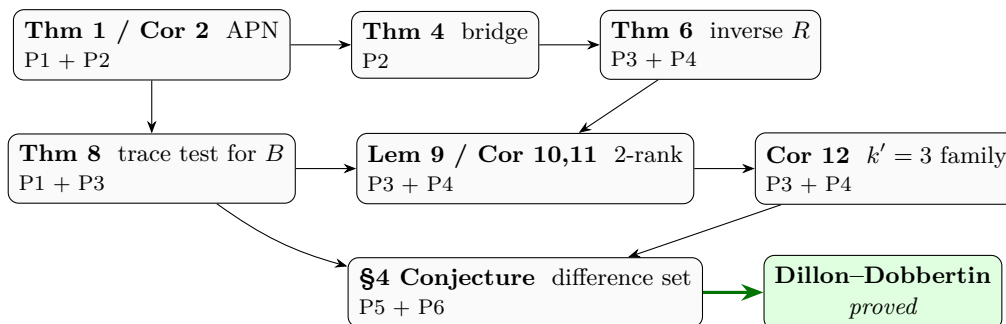
✿ IN THE LEAN PROJECT

The project records the uniform version as `WidthUnif` and proves the second-moment identity `coll_of_uniform` ($\sum_b \text{wid}^2 = \omega \cdot |S|$) — the algebraic signature of “flat”.

```
def WidthUnif ω( d : ℕ) : Prop :=
  ∀ a, a ≠ S.unit_ → ∀ b, wid S d a b = 0 ∨ wid S d a b = ω
theorem coll_of_uniform (d ω : ℕ) (ωh : 0 < ω) (hu : WidthUnif S ω d) ... :
  ∑ b, wid S d a b ^ 2 = ω * Fintype.card S.obj
```

How the patterns propagate

Each result is a *combination* of the six shapes. Follow the coloured tags.



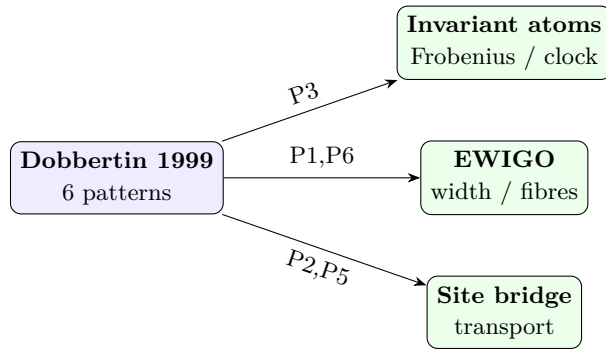
step	patterns fired
Kasami is a permutation, hence APN	P1 (two-to-one) + P2 (bridge to MCM)
Explicit inverse R	P3 (clock) + P4 (Fibonacci recursion)
$x \in B \iff \text{Tr}(R(x)) = 0$	P1 (fibres) + P3 (trace / cosets)
2-rank and Segre / $k' = 3$ families	P3 (necklaces) + P4 (Fibonacci)
B^* is a cyclic difference set	P5 (quadratic echo) + P6 (flat/ideal)

★ BIG IDEA

The paper is a *relay race*. P1 starts it (count fibres); P2 duplicates the lane (Kasami $\equiv$ MCM); P3+P4 build an explicit, sparse inverse; that inverse turns the geometric set B into a trace test; and P5+P6 hand the baton to a difference-set conjecture — later crossed the finish line by Dillon & Dobbertin.

Where the patterns live in the Lean project

pattern	Dobbertin object	Lean declaration / file
P1	two-to-one map; APN; width 2	tDiff, fib, wid, WidthBdd (EWIGO)
P2	Thm 4 Kasami $\leftrightarrow$ MCM	SiteEquiv, Pattern.transport, width_transport (Kasami/SiteEquivalences.lean)
P3	cyclotomic cosets, reduced form	pow_invariant_mod, frob_bij (Kasami/InvariantPatterns.lean)
P4	recursions A_i, B_i, C_i ; $w(R)$	Fibonacci counting, Nat.fib
P5	quadratic vs. Kasami exponent	Gold vs. Kasami power maps
P6	ideal autocorrelation, difference set	WidthUnif, coll_of_uniform; flat-spectrum reading



One sentence to remember

☞ Count fibres (P1), copy across a bridge (P2), work on the exponent clock (P3), let Fibonacci do the counting (P4), echo the quadratic case (P5), and check for flatness (P6)
 — that is the whole Dobbertin paper.